11 Chemistry — Unit 2 Topic Test 4: Gases II (Gas Stoichiometry)

SOLUTIONS AND MARKING GUIDE — teacher copy; release to students only after the test has been sat

Total 40 marks: Section A 10 × 1 mark (separate answer sheet), Section B 30 marks. Writing time 46 min (40 marks × 1.15 min per mark). Students use the VCE Chemistry Data Book (SLC 25 °C and 100 kPa; V_m = 24.8 L mol^{-1} at SLC; R = 8.31 J mol^{-1} K^{-1}; 1 atm = 101.3 kPa; c(water) = 4.18 J g^{-1} °C^{-1}; T(K) = T(°C) + 273). **One mark per idea or step; no half marks.** Mark units and significant figures **once per part, at the final answer** (3 significant figures expected unless stated; accept 2–4 where the data allow). **Rounding tolerance:** accept a final answer within ±1 in the last significant figure when it comes from correctly rounded intermediate values; each part states its accepted range. **Consequential marking:** where a later part uses an earlier answer, award full marks for correct method applied to the student's earlier value; parts where this applies are flagged *Consequential*. Energy released or absorbed is a positive magnitude. Equations need correct states.

Section A — Multiple-choice answer key (10 marks)

Key: B D A C A D B C A B. Balance: A ×3, B ×3, C ×2, D ×2; no letter more than 1 in a row. Each wrong letter points to one slip or misconception (right-hand column). Difficulty and demand level for each item are shown in italics after the working.

Q	Key	Working for the key	Distractor diagnosis
1	B	n(Ar) = 1.50 ÷ 24.8 = 0.0605 mol; m = 0.0605 × 39.9 = 2.41 g. *[Easy; L2]*	**A** used 22.4 L mol^{-1} (0 °C molar volume) instead of 24.8 L mol^{-1} at SLC; **C** used 18 (atomic number) as the molar mass of argon instead of 39.9 g mol^{-1}; **D** stopped at the amount in mol and reported it as a mass
2	D	Equal volumes of gases at the same T and p contain equal amounts (mol), so equal numbers of molecules (Avogadro). *[Easy; L1]*	**A** thinks equal volumes have equal masses (molar masses 16.0 and 44.0 differ); **B** confuses molecules with atoms (CH_4 has 5 atoms per molecule, CO_2 has 3); **C** density = mass ÷ volume; the masses differ, so the densities differ
3	A	Definition: SLC is 25 °C and 100 kPa; V_m applies to 1 mol of any (ideal) gas. *[Easy; L1]*	**B** confuses amount (mol) with mass; **C** wrong temperature: SLC is 25 °C (298 K); **D** thinks molar volume depends on the identity of the gas
4	C	n(CaH_2) = 10.5 ÷ 42.1 = 0.249 mol; n(H_2) = 0.499 mol; V = nRT ÷ p = 0.499 × 8.31 × 288 ÷ 101 = 11.8 L. *[Medium; L2]*	**A** used a 1:1 mole ratio instead of 1:2; **B** used T = 15.0 (°C) instead of 288 K; **D** used V_m = 24.8 L mol^{-1}, which applies only at SLC
5	A	At the same T and p, volume ratio = mole ratio: V(O_2) = 5 × 2.0 = 10.0 L. *[Easy; L2]*	**B** assumed a 1:1 volume ratio; **C** gave the volume of CO_2 formed, not O_2 used; **D** inverted the 1:5 ratio
6	D	At the same T and p, volume ratio = mole ratio. 80 mL O_2 reacts with 16 mL propane (O_2 is limiting), forming 48 mL CO_2; 4 mL propane is left; water is liquid. Total 52 mL. *[Hard; L3]*	**A** counted the CO_2 formed but forgot the propane left over; **B** counted the water as a gas (48 + 64 + 4 mL); **C** assumed the total volume of gas does not change in a reaction
7	B	V_m = RT ÷ p: 100 °C, 100 kPa gives 8.31 × 373 ÷ 100 = 31.0 L mol^{-1}, the largest. *[Medium; L3]*	**A** thinks a higher pressure gives a larger volume (V_m = 12.4 L mol^{-1}); **C** thinks colder gas occupies more volume (V_m = 22.7 L mol^{-1}); **D** chose SLC by habit (V_m = 24.8 L mol^{-1})
8	C	n(N_2) = 1.50 ÷ 24.8 = 0.0605 mol = n(NH_4NO_2); m = 0.0605 × 64.0 = 3.87 g. *[Medium; L2]*	**A** used the coefficient of H_2O (2) in the mole ratio; **B** used 22.4 L mol^{-1} instead of 24.8 L mol^{-1}; **D** multiplied by the molar mass of N_2 (28.0) instead of NH_4NO_2 (64.0)
9	A	n(Al) = 0.270 ÷ 27.0 = 0.0100 mol; n(NaOH) = 0.500 × 0.0150 = 7.50 × 10^{-3} mol. Ratio 1:1, so NaOH is limiting. n(H_2) = 1.5 × 7.50 × 10^{-3} = 0.01125 mol; V = 0.01125 × 24.8 = 0.279 L. *[Hard;*	**B** assumed aluminium is the limiting reactant (it is in excess); **C** used a 1:1 ratio for NaOH : H_2; **D** used a 1:3 ratio for NaOH : H_2 instead of 2:3

		L3]	
10	B	$n(CO) = 0.140 \div 28.0 = 5.00 \times 10^{-3}$ mol = $n(CO_2)$; $V = 5.00 \times 10^{-3} \times 24.8 = 0.124$ L = 124 mL. *[Medium; L2]*	**A** used a 2:1 ratio for CO : CO_2 (halved the amount); **C** doubled the amount of CO (used a 1:2 ratio); **D** multiplied the mass of CO by V_m without converting to mol

Section B — Worked solutions and marking scheme (30 marks)

Each part shows the full worked solution, then the bold mark allocation (one mark per idea), then *Consequential* and *Reject* notes. Difficulty and demand level are shown in italics.

Question 1 — Carbide lamps (7 marks)

a) (1 mark)

$M(CaC_2) = 40.1 + 2 \times 12.0 = 64.1$ g mol^{-1}

$n(CaC_2) = m \div M = 40.0 \div 64.1 = 0.624$ mol

1 mark 0.624 mol (accept 0.62–0.624). *[Easy; L2]*

Reject: Using M(Ca) only (0.998 mol).

b) (2 marks)

Mole ratio $CaC_2 : C_2H_2 = 1 : 1$, so $n(C_2H_2) = 0.624$ mol

$V = n \times V_m = 0.624 \times 24.8 = 15.5$ L

1 mark $n(C_2H_2) = 0.624$ mol from the 1:1 ratio. **1 mark** V = 15.5 L (accept 15.4–15.5 L; unit required). *[Easy; L2]*

Consequential: On the amount from part a).

Reject: 22.4 L mol^{-1} (14.0 L).

c) (2 marks)

$T = 14.0 + 273 = 287$ K; $n(C_2H_2) = pV \div RT = (99.0 \times 30.0) \div (8.31 \times 287) = 1.245$ mol

$n(CaC_2) = n(C_2H_2) = 1.245$ mol; $m = 1.245 \times 64.1 = 79.8$ g

1 mark $n(C_2H_2) = pV \div RT = 1.25$ mol (T in K, p in kPa, V in L). **1 mark** $m(CaC_2) = 79.8$ g (accept 79.7–80.0 g; unit required). *[Medium; L2]*

Reject: Using 24.8 L mol^{-1} at cave conditions (77.5 g): first mark lost. Using T = 14.0 (°C).

d) (2 marks)

The lower chamber holds far more CaC_2 than the water that has dripped in can react with, so CaC_2 is in excess and water is the limiting reactant.

The amount of ethyne formed depends on the amount of water added (2 mol H_2O → 1 mol C_2H_2), so less water per minute gives less ethyne per minute at the jet, and a smaller flame.

1 mark identifies water as the limiting reactant (CaC_2 in excess). **1 mark** links less water added per minute to less C_2H_2 produced per minute (via the mole ratio), so less gas burns. *[Medium; L3]*

Reject: "Calcium carbide is the limiting reactant." "Slowing the drip slows the reaction" with no limiting-reactant reasoning (0 for the first mark).

Question 2 — Molar mass of the gas from an antiseptic (9 marks)

a) (2 marks)

$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

1 mark correct formulas: $H_2O_2 \rightarrow H_2O + O_2$. **1 mark** balanced (2 : 2 : 1) with all states correct. MnO_2 written over the arrow is accepted, not required. *[Medium; L2]*

Reject: H_2 as the gas. O as a product. $H_2O_2 \rightarrow H_2 + O_2$.

b) (3 marks)

Experiment 2: p = 98.5 kPa, V = 0.438 L, T = 21.0 + 273 = 294 K

$n = pV \div RT = (98.5 \times 0.438) \div (8.31 \times 294) = 0.01766$ mol (1.77×10^{-2} mol)

Experiment 1: mass of gas = mass lost = 0.566 g

$M = m \div n = 0.566 \div 0.01766 = 32.1$ g mol^{-1}

1 mark $n = pV \div RT$ with V in L and T in K. **1 mark** n = 0.0177 mol (accept 0.0176–0.0177). **1 mark** $M = m \div n$ using 0.566 g: 32.1 g mol^{-1} (accept 31.9–32.2; unit required). *[Medium; L3]*

Consequential: Final mark on the student's n.

Reject: V in mL (n = 17.7 mol, M = 0.0321). n from 24.8 L mol^{-1} (not SLC). 32.0 from the formula alone with no calculation (0).

c) (2 marks)

$n(H_2O_2) = 2 \times n(O_2) = 2 \times 0.01766 = 0.03533$ mol

$m(H_2O_2) = 0.03533 \times 34.0 = 1.20$ g in 40.0 mL

% (m/v) = 1.20 g ÷ 40.0 mL × 100 = 3.00 % (m/v), which matches the label.

1 mark $n(H_2O_2) = 2 \times n(O_2) = 0.0353$ mol and m = 1.20 g. **1 mark** 3.00 % (m/v) (accept 2.99–3.01 %; accept 3.0 %). *[Hard; L2]*

Consequential: On the equation in part a) and n from part b). Using $n(O_2) = 0.566 \div 32.0$ also scores (3.01 %).

Reject: Ratio 1:1 (1.50 %). Dividing by 40.0 without × 100 (0.0300). Mass of O_2 used as the mass of H_2O_2 (1.42 %).

d) (2 marks)

The decrease in mass would include the escaped droplets as well as the gas, so the measured "mass of gas" (0.566 g) is too high.

The amount of gas (from Experiment 2) is unaffected, so $M = m \div n$ is calculated too high.

1 mark the mass loss is too large / includes the droplets, so the mass of gas used is too high. **1 mark** n is unchanged, so the calculated molar mass is too high. *[Hard; L4]*

Reject: "Lower." "No effect, because the volume of gas is the same." "Less accurate" with no direction (0). Answers about gas volume or temperature.

Parts b) and c) form an open chain; c) is consequential on a) and b). Part d) is an error-direction item: only answers that push the molar mass up score.

Question 3 — Copper and nitric acid (7 marks)

a) (1 mark)

$n(Cu) = 1.27 \div 63.5 = 0.0200$ mol

1 mark 0.0200 mol. *[Easy; L2]*

b) (2 marks)

$n(HNO_3) = c \times V = 12.0 \times 0.00500 = 0.0600$ mol

To react with 0.0200 mol Cu needs $4 \times 0.0200 = 0.0800$ mol HNO_3; only 0.0600 mol is present.

So nitric acid is the limiting reactant (copper is in excess).

1 mark $n(HNO_3) = 0.0600$ mol. **1 mark** uses the 1:4 ratio to compare (0.0800 mol needed > 0.0600 mol, or $0.0600 \div 4 = 0.0150$ mol < 0.0200 mol) and concludes HNO_3 is limiting. *[Medium; L3]*

Consequential: On n(Cu) from part a).

Reject: Concluding from amounts without the 1:4 ratio (0.0600 > 0.0200, so copper is limiting): 0 for the second mark.

c) (2 marks)

$n(NO_2) = ½ \times n(HNO_3) = ½ \times 0.0600 = 0.0300$ mol (ratio 4 : 2)

$V = 0.0300 \times 24.8 = 0.744$ L (744 mL)

1 mark n(NO_2) = 0.0300 mol from the limiting reactant. **1 mark** 0.744 L (or 744 mL; unit required). *[Medium; L2]*

Consequential: On the limiting reactant chosen in part b): a student who chose copper in b) and gives 2 × 0.0200 × 24.8 = 0.992 L scores 2 marks here; a student who chose HNO_3 in b) but then uses copper scores 1 mark.

Reject: Using 22.4 L mol^{-1} (0.672 L). Ratio 1:1 for HNO_3 : NO_2 (1.49 L).

d) (2 marks)

n(HNO_3) needed = 4 × 0.0200 = 0.0800 mol; present = 0.0600 mol; extra = 0.0200 mol

V = n ÷ c = 0.0200 ÷ 12.0 = 1.67 × 10^{-3} L = 1.67 mL

1 mark extra amount = 0.0200 mol (or total volume needed 6.67 mL). **1 mark** extra volume 1.67 mL (accept 1.66–1.67 mL). *[Hard; L3]*

Consequential: On parts a) and b).

Reject: 6.67 mL (total, not the extra). 0.00167 mL.

Question 4 — Baker's ammonia (7 marks)

a) (1 mark)

M(NH_4HCO_3) = 14.0 + 5(1.0) + 12.0 + 3(16.0) = 79.0 g mol^{-1}

n = 3.16 ÷ 79.0 = 0.0400 mol

1 mark 0.0400 mol. *[Easy; L2]*

b) (3 marks)

Each mol of NH_4HCO_3 gives 3 mol of gas: n(gas) = 3 × 0.0400 = 0.120 mol

T = 180 + 273 = 453 K

V = nRT ÷ p = 0.120 × 8.31 × 453 ÷ 100 = 4.52 L

1 mark n(gas) = 0.120 mol. **1 mark** V = nRT ÷ p with T = 453 K. **1 mark** 4.52 L (accept 4.51–4.52 L; unit required). *[Medium; L2]*

Consequential: On n from part a).

Reject: One gas only (1.51 L). V_m = 24.8 L mol^{-1} at oven temperature (2.98 L). T = 180 (1.79 L).

c) (2 marks)

At 25 °C, water is a liquid, so only NH_3 and CO_2 remain as gases: n(gas) = 2 × 0.0400 = 0.0800 mol

V = 0.0800 × 24.8 = 1.98 L

1 mark recognises that water condenses at SLC, so n(gas) = 0.0800 mol. **1 mark** 1.98 L. *[Hard; L3]*

Consequential: On n from part a).

Reject: 2.98 L (water still counted as a gas). 4.52 L (no change). 0.99 L (one gas only).

d) (1 mark)

n(NH_3) = n(NH_4HCO_3) = 0.0400 mol; m = 0.0400 × 17.0 = 0.680 g

1 mark 0.680 g (unit required). *[Easy; L2]*

Consequential: On n from part a).

Reject: 2.04 g (used 3 × 0.0400 mol). 0.0400 g.

Mark summary

Section	Marks	Parts
A — multiple choice	10	10 × 1 mark; key A3 / B3 / C2 / D2
B Q1 — Carbide lamps	7	a 1, b 2, c 2, d 2
B Q2 — Molar mass of the gas from an antiseptic	9	a 2, b 3, c 2, d 2
B Q3 — Copper and nitric acid	7	a 1, b 2, c 2, d 2

B Q4 — Baker's ammonia	7	a 1, b 3, c 2, d 1
Total	40	40 marks × 1.15 min = 46 min writing time

Difficulty tally (marks)

Target for every topic test: 25% easy (10 marks), 50% medium (20 marks), 25% hard (10 marks).

	Easy	Medium	Hard	Total
Section A (MC)	4	4	2	10
Section B (short answer)	6	16	8	30
Whole test (marks)	10	20	10	40
Share of marks	25%	50%	25%	100%

Difficulty and demand by item

Item	Marks	Difficulty	Demand
A1	1	Easy	L2
A2	1	Easy	L1
A3	1	Easy	L1
A4	1	Medium	L2
A5	1	Easy	L2
A6	1	Hard	L3
A7	1	Medium	L3
A8	1	Medium	L2
A9	1	Hard	L3
A10	1	Medium	L2
B1a	1	Easy	L2
B1b	2	Easy	L2
B1c	2	Medium	L2
B1d	2	Medium	L3
B2a	2	Medium	L2
B2b	3	Medium	L3
B2c	2	Hard	L2
B2d	2	Hard	L4
B3a	1	Easy	L2
B3b	2	Medium	L3
B3c	2	Medium	L2
B3d	2	Hard	L3
B4a	1	Easy	L2
B4b	3	Medium	L2
B4c	2	Hard	L3
B4d	1	Easy	L2

Cognitive demand

Demand level	Marks	Share
L1 Recall / identify	2	5%
L2 Routine procedure	22	55%
L3 Interpret / explain	14	35%
L4 Evaluate / error direction	2	5%

Study design coverage

Key knowledge (Unit 2)	Where assessed
Stoichiometry of reactions involving gases: mass ↔ mol ↔ volume at SLC	A1, A5, A6, A8, A10; B1b, B3c, B4c, B4d
pV = nRT with unit conversions (kPa, mL, °C → K)	A4, A7; B1c, B2b, B4b
Molar volume (meaning, dependence on conditions)	A2, A3, A7
Molar mass of a gas produced by a reaction	B2a, B2b, B2d

Limiting and excess reactants	A6, A9; B1d, B3b, B3d
Concentration (mol L^{-1}, % (m/v)) linked to gas data	A9; B2c, B3b, B3d